

Forms and Their Styling

Lesson 10



Lesson plan

1

HTML Form Elements

2

Styling Form Elements

3

Challenges with Styling

4

Form Validation

5

Other Interactive Tags

Registration

Full Name

E.g: John Smith

Username

johnWC98

Email

johnsmith@hotmail.com

Phone Number

012-345-6789

Password

Confirm Password

Gender

☐ Male

☐ Female

☐ Prefer not to say

Register

Send us a Message



CodingNepal



Enter your email



Enter your phone



Enter your website



Write your message

Send Message

Sending your message...

Price Range

Use slider or enter min and max price

Min

2500

-

Max

7500

Subscribe to the newsletter

your@email.com

Subscribe



Start typing in here to take over and try it out

5423

Jondoe Donjoe

12 / 12

121

Submit

Upload File



Click To Upload

Uploaded Documents

Pdf

document.folder.pdf



Pdf

Resume.pdf



Docx

Ravi Web.docx



Start Date

04-05-2020



End Date

22-05-2020



May 2020						
	Su	Mo	Tu	We	Th	Fr
18	26	27	28	29	30	1
19	3	4	5	6	7	8
20	10	11	12	13	14	15
21	17	18	19	20	21	22
22	24	25	26	27	28	29
23	31	1	2	3	4	5

Clear

INGREDIENTS



Beef



Pork



Chicken



Vegetables

NUTRITION



Gluten Free

Your email address:

test@example.org

Interactive form demo

Test message *

Optional files

Browse...

No files selected.

Send to me

Demo source code

```
<!-- source HTML for the demo -->
<form
  action="https://api.mailslurp.com/forms"
  method="post"
  enctype="multipart/form-data">

  <!-- destination specified with hidden '_to' fi
  <input
    name="_to"
    type="hidden"
    value="test@example.org">

  <!-- use any named inputs, selects, textareas e
  <label>Test message</label>
  <textarea name="message"><textarea>

  <!-- files will be sent as attachments -->
  <label>Optional files</label>
  <input multiple name="files" type="file">

  <button type="submit">Send to me</button>

</form>
```

<form> Attributes



action – The **URL** where the form data will be sent

method – The HTTP method used for sending form data

GET – Used to retrieve data

POST – Used to send data, especially sensitive information

```
<form action="submit.php" method="post">
```

Grouping Elements

- **<fieldset>** – Groups related elements within a form
- **<legend>** – A caption for the fieldset
 - The caption is offset upwards, which can make it difficult to align with other elements
 - It does not respond to **flex** layout properties applied to the **fieldset**

Input text

Login Id:

Password:

Employee Designation

☒ Software Engineer

☐ Data Analyst

☐ Web Developer

☐ Senior Analyst

press Reset



EVENT Registration

Friday 15th & Saturday 16th February @ 8pm

YOUR DETAILS

Name		Country	
Phone		State	
Email		Mem No.	

TICKETS

Date	☐ Fri ☐ Sat	Delivery	☐ Email ☐ Collect
Quantity		Promotional Code	

Deadline for Registration is Friday 4th January

eTickets will be issued to your email address two weeks before the event.
Gates will open three hours before the event gets started.
Please remember to bring identification if collecting your ticket from our events desk.

LET'S GET THIS
**PARTY
STARTED**

SUBMIT

User settings

Lorem ipsum dolor sit amet, consectetur.

Personal info

First Name		Last Name	
Country	v		

Notifications

Weekly reports	☐
Pull requests	☐
Deployment triggers	☐

Security

Run security check upon log in ☒

Less More

Save **Cancel**

Tell us about your home.

Address

YOUR INFORMATION IS PRIVATE

Address

Address 2

City City	State State	Zip Code Zip Code
--	--	--

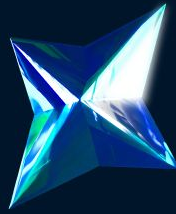
Bed & Bath

WHAT IS YOUR HOME LIKE?

Home Type House	Room Type Entire home/apt		
Bedrooms 1 Bedroom	Bathrooms 1 Bathroom		
King 0	Queen 0	Full 0	Twin 0

Sign me up!

<input>



- Used to create interactive form controls for user data input.
- The default type is **text**
- One of the most powerful and complex HTML elements due to the vast number of input types and attributes.

<input type="text">



input attributes

– customize `<input>`

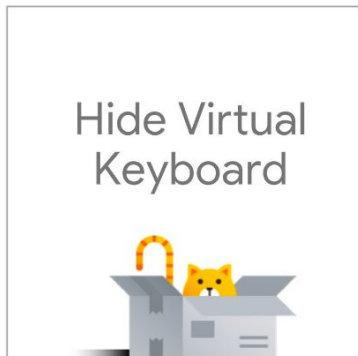
- These attributes add features and behaviors to the `<input>` elements.
- Attribute examples include **size**, **value**, **maxlength**, **required**, and many more.

```
<input  
  type="text"  
  name="bro"  
  required  
  minlength="2"  
  maxlength="8"  
/>
```

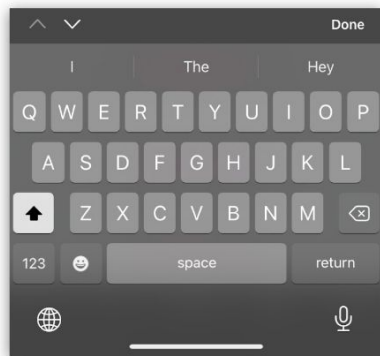
input types

<code><input type="button"></code>	Button
<code><input type="checkbox"></code>	☐ ☒ ☐
<code><input type="color"></code>	<input type="color" value="#ff0000"/>
<code><input type="date"></code>	<input type="date" value="dd-mm-yyyy"/> 
<code><input type="email"></code>	<input type="email" value="farazc60@gmail.com"/>
<code><input type="file"></code>	<input type="file"/> Choose File No file chosen
<code><input type="hidden"></code>	
<code><input type="image"></code>	<input type="image" value="img"/>
<code><input type="number"></code>	<input type="number" value="5"/>
<code><input type="password"></code>	
<code><input type="radio"></code>	☐ ☐
<code><input type="range"></code>	<input type="range" value="50"/>
<code><input type="reset"></code>	<input type="reset" value="Reset"/>
<code><input type="submit"></code>	<input type="submit" value="Submit"/>
<code><input type="text"></code>	codewithfaraz
<code><input type="url"></code>	<input type="url" value="https://www.codewithfaraz.com"/>

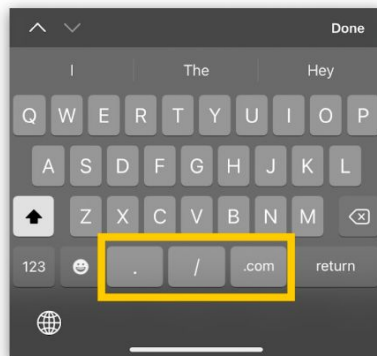
<input inputmode="..">



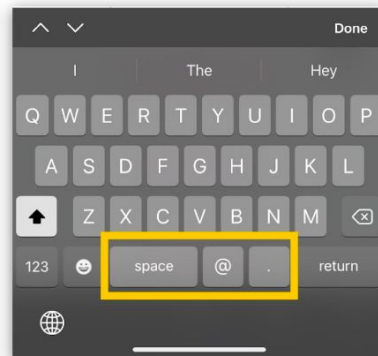
"none"



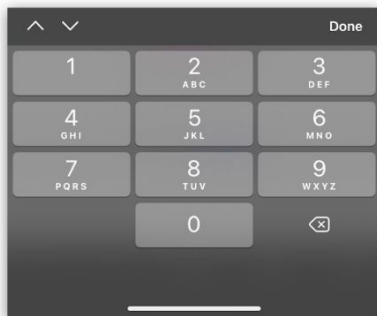
"text"



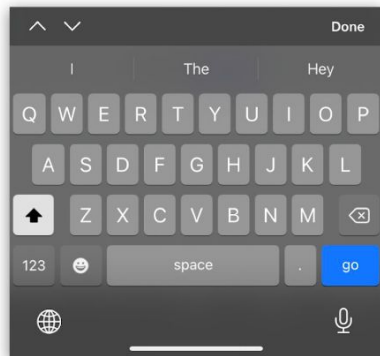
"url"



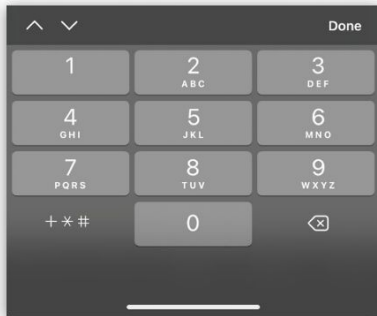
"email"



"numeric"



"search"



"tel"



"decimal"

label

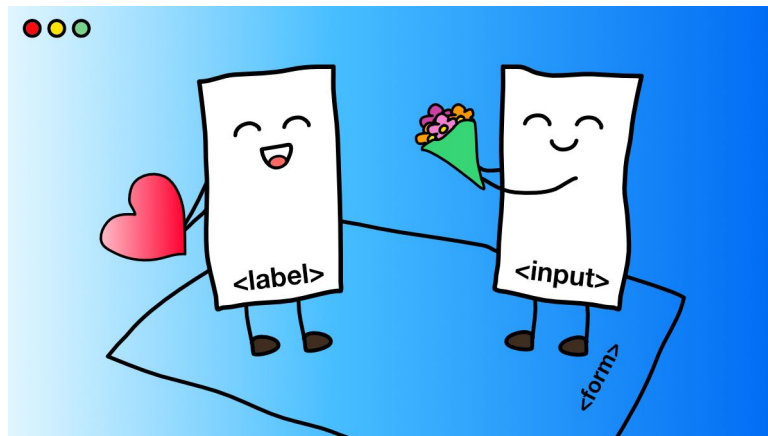
- description for a form element
- linked using the **id** attribute
- clicking on the **label** focuses on the associated form element
- !every form element should have a corresponding label
- can be used for styling elements that are otherwise difficult to style

Label

input

Name

john wick



```
<input type="radio" name="vegetables" value="tomato"
id="vegetables-tomato">
<label for="vegetables-tomato"> Tomato </label>
```

Checkbox Usage Rules

– used to let a user select **one or more** options of a limited number of choices.

```
<input type="checkbox" name="vegetables" value="tomato"  
class="radio visually-hidden" id="checkbox-tomato">
```

Required Attributes:

value – Must be unique for each checkbox

Optional:

name – Should be the same for all checkboxes in a group



Tomato



Onion



Lettuce



Capcicum

Radio Button Usage Rules

– used to let a user select **one** option of a limited number of choices.

```
<input type="radio" name="vegetables" value="onion" class="radio  
visually-hidden" id="vegetables-onion">
```

Required Attributes:

name – Required Attributes

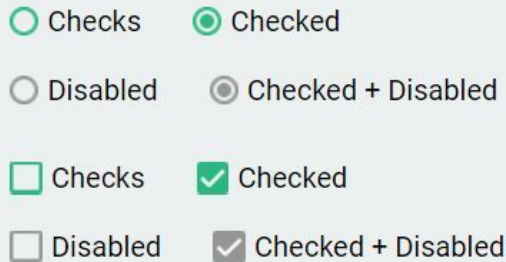
value – Must be unique for each radio button

☐ **Tomato**

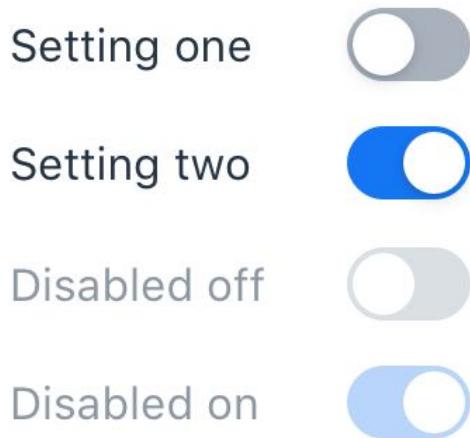
☒ **Onion**

☐ **Lettuce**

☐ **Capcicum**



Toggle button



`<input>`

radio & checkbox

- radio and checkbox elements cannot be easily styled directly
(though it is possible)
- Typically, the `input` element is hidden, and the associated `label` is styled instead
- Always consider the `checked` and `disabled` states for styling

Styling Radio & Checkbox **:checked**

```
<input type="radio" name="weight" value="weight0" class="radio  
visually-hidden" id="weight0">
```

```
<label for="weight0">0%</label>
```

Styles for Unchecked Elements:

```
.radio:not(:checked) + label {}
```



Styles for Checked Elements:

```
.radio:checked + label {}
```


Accessible **input** hiding

```
<input type="radio" class="visually-hidden">
```

~~display: none;~~

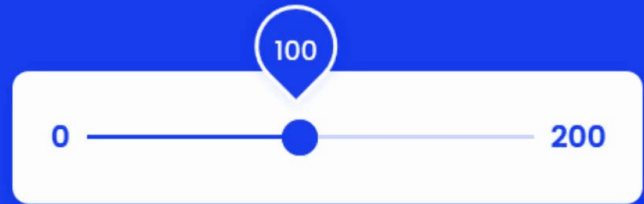
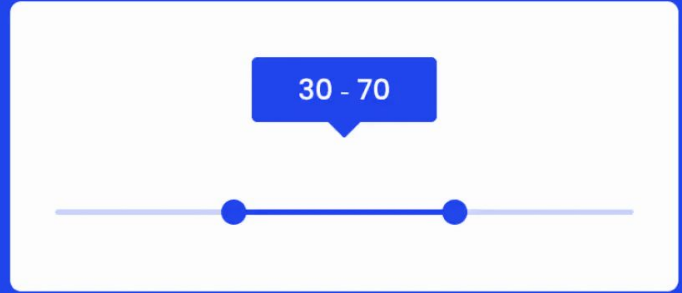
```
.visually-hidden {  
  
    position: absolute;  
    height: 0;  
    width: 0;  
    opacity: 0; /* or: visibility: hidden; */  
  
}
```

input

range

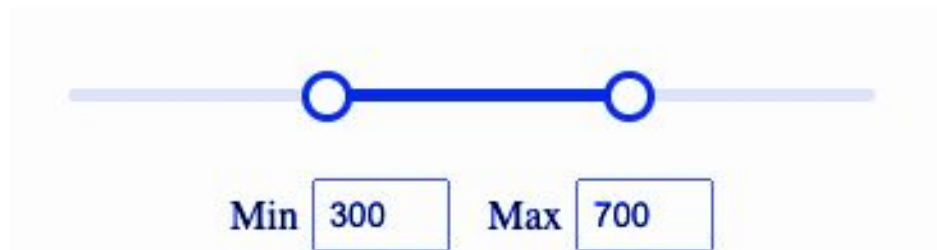
- Has a single slider
- Difficult to style
- usually requires a plugin for more complex controls

For styling, consider using a JS plugin like [AlRangeSlider](#).



input range in HTML

```
<div class="range">
  <button class="range-button is-min"></button>
  <button class="range-button is-max"></button>
  <div class="range-line"></div>
  <div class="range-line is-selected"></div>
  <div class="range-container">
    <div class="range-min"> Min
      <input type="number">
    </div>
    <div class="range-max"> Max
      <input type="number">
    </div>
  </div>
</div>
```

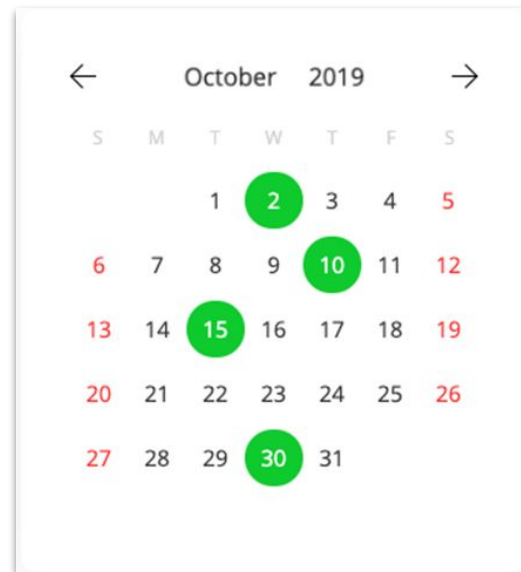
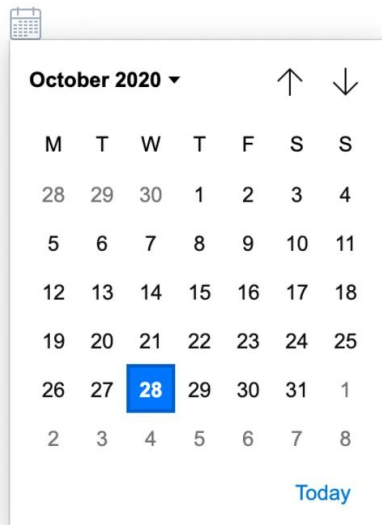


styling datepicker

`<input type="date">`

— Cannot be styled natively.

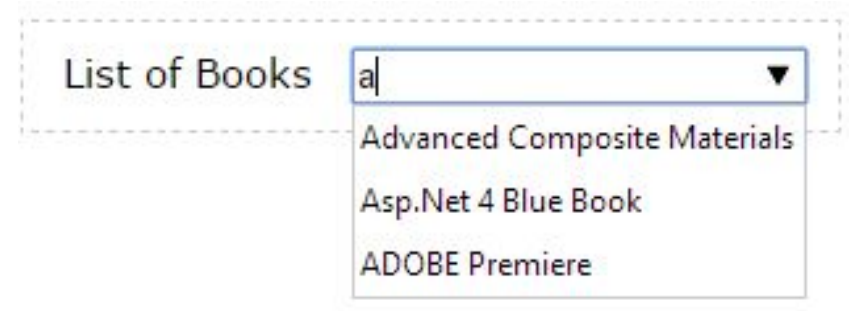
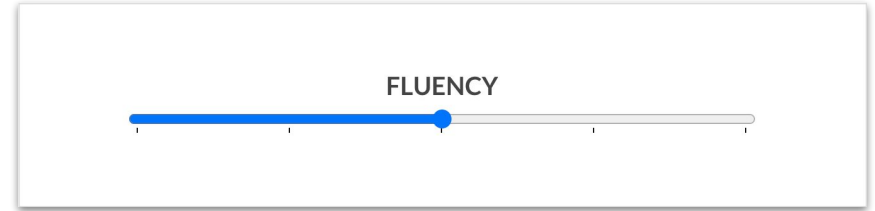
— Use JS plugins like [Air datepicker](#) for custom styling



datalist

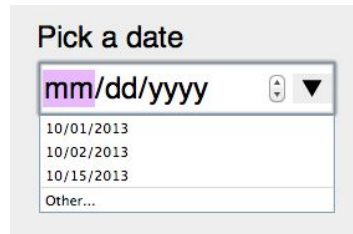
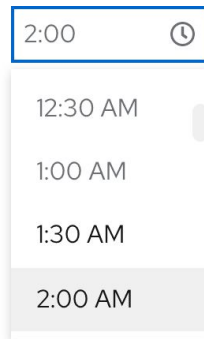
- Provides a list of options for an **input** element.
- You can select an option from the list instead of entering it manually
- Works with specific input types such as:

**text, search, range, color,
number, date, time, tel, email**



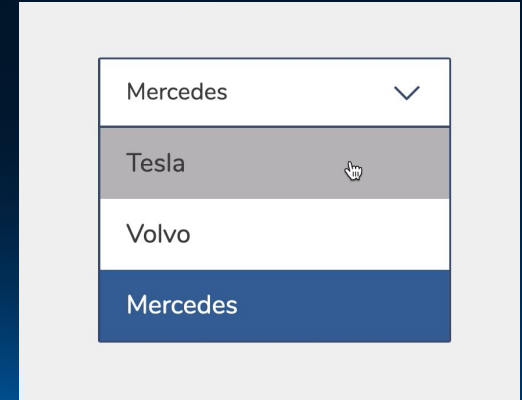
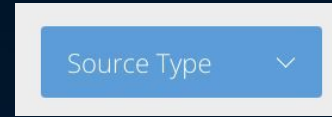
datalist

```
<input list="options" name="input_name">
<datalist id="options">
  <option value="option 1">
  <option value="option 2">
  <option value="option 3">
</datalist>
```



select

- Dropdown menu allowing the selection of one or multiple options
- The dropdown button can be partially styled, but not the arrow
- The dropdown cannot be natively styled



select

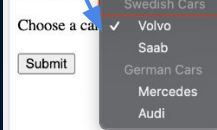
```
<select>
  <option value="apple">Apple</option>
  <option value="orange">Orange</option>
  <option value="berry">Berry</option>
</select>
```

- option** – an element in a dropdown list
 - value** – the value of the element that will be sent to the server; a required attribute
- optgroup** – an element used to group items in a dropdown list
 - label** – an attribute that sets the title for a group (the title is not clickable)

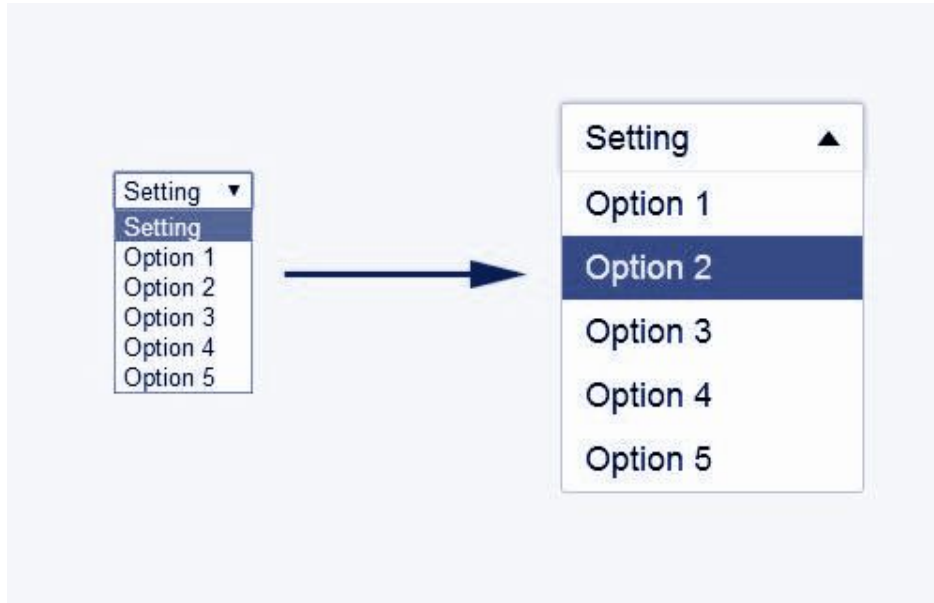
```
<form action="/action_page.php">
  <label for="cars">Choose a car:</label>
  <select name="cars" id="cars">
    <optgroup label="Swedish Cars">
      <option value="volvo">Volvo</option>
      <option value="saab">Saab</option>
    </optgroup>
    <optgroup label="German Cars">
      <option value="mercedes">Mercedes</option>
      <option value="audi">Audi</option>
    </optgroup>
  </select>
  <br><br>
  <input type="submit" value="Submit">
</form>
```

The optgroup element

The optgroup element is used to group related options in a drop-down list:



styling **select**



Using JavaScript libraries.

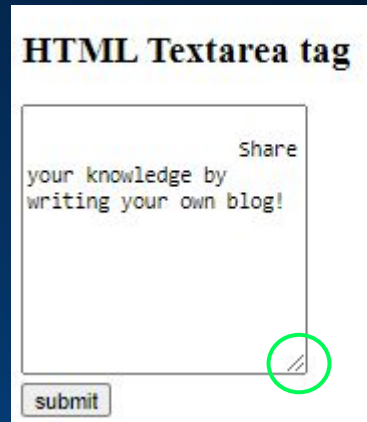
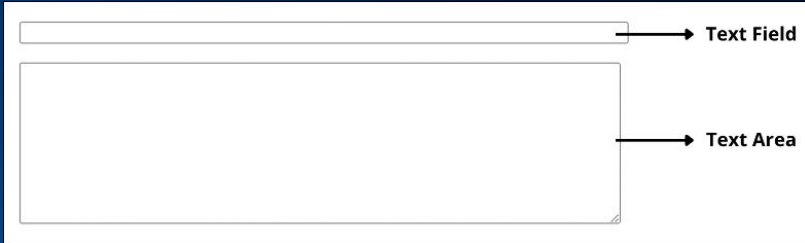
They create a separate element next to the **<select>** for styling purposes, while hiding the original **<select>** element.

When an option is selected, the choice is passed to the hidden **<select>**, which is then submitted with the form.

For example, **jQuery Nice Select**.

textarea

- Allows multi-line text input
- By default, it is resizable. This can be disabled using CSS



textarea

Attributes:

cols – specifies the width of the textarea in terms of the number of text columns

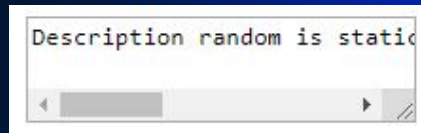
rows – specifies the number of text lines it can display at a time (height)

instead of using these attributes, you might set the width and height in your CSS

if you want to prevent users from resizing:

```
textarea {  
    resize: none;  
}
```

```
<div class="form-row">  
  <textarea name="basic-textarea" id="basic-textarea"  
    class="form-control" cols="30" rows="10"></textarea>  
  <label for="basic-textarea">Basic Textarea</label>  
</div>
```



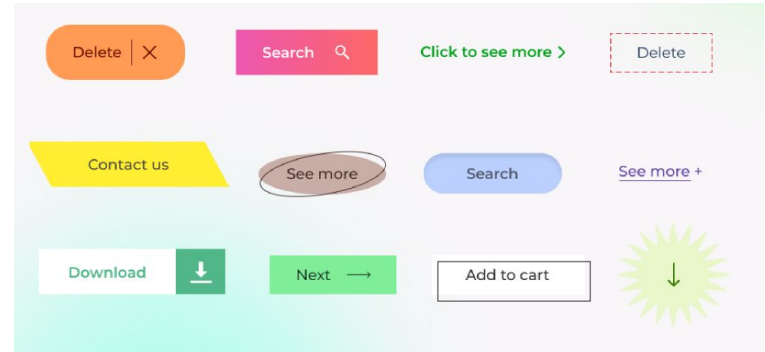
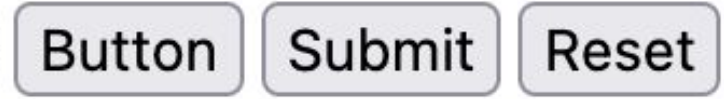
button

button – A generic button for any action

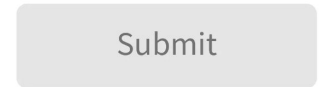
submit – Submits the form data to the server

reset – Resets the form data to its initial state

The **disabled** attribute can be used to prevent the form from being submitted



Button Enabled



Button Disabled



What is your rate?



Please share your opinion
about the product



Drop Us A Message

Name

Email Address

Write Us A Message

Send

Sign Up Sign In

Email

Username

Password

SIGN UP

Enter your mail

Subscribe

Captcha:

Form

☒ Pizza ☐ Pasta

[Reset form](#)

00 : 00 : 00 : 000

form validation

- Validates form data according to specific rules

- Methods for validation include:

- `html`
- `js`

test

Last name

test@test.com

.....

An account has already been registered with this email. [Log in](#) to your account.

Sign Up

Sign Up

Username:

js

Username must be between 3 and 25 characters.

Email:

hello@example.com

Password:

...

Password must have at least 8 characters that include at least 1 lowercase character, 1 uppercase character, 1 number, and 1 special character in (!@#\$%^&*)

Confirm Password:

Reenter your password

Please enter the password again

SIGN UP

HTML form validation

1. Use the appropriate **input type**
2. Add the **required** attribute where necessary

```
<input type="email" required>
```

3. Use the **pattern** attribute for custom validation rules
4. Style inputs and validation messages using pseudo-classes like **:invalid**, **:valid**

input attribute **pattern**

- helps ensure data is entered in the correct format
- provides a prompt or error message if the data doesn't match the pattern
- ask ChatGPT to help you create the right pattern

Use sites like <https://regex101.com> for **pattern** checking.

pattern=" [A-Za-z]{5}"

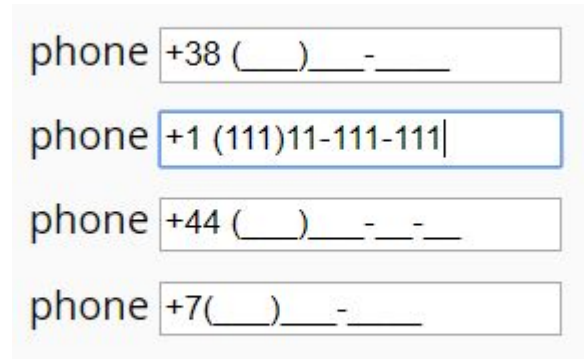
This pattern requires 5 characters, allowing both uppercase and lowercase Latin letters.



USERNAME

john1

Please match the requested format.
Username should only contain lowercase letters. e.g. john



phone +38 () -

phone +1 (111)11-111-111|

phone +44 () - -

phone +7() -

```
<input type="email" pattern="[a-z0-9._%+-]+@[a-z0-9.-]+\.[a-z]{2,}$">
```


User interface **pseudo-class** selectors

- **:required** – targets elements that have the **required attribute** (indicating that the field is mandatory)
 - **:optional** – targets elements that **do not have** the required attribute
-



User interface **pseudo-class** selectors

- **:disabled** – element that is disabled (using the disabled attribute)
- **:enabled** – elements that **do not have** the disabled attribute
- **:default** – the element that is the **default** among a group of similar elements

Default

Label here *

Disabled

Label here *

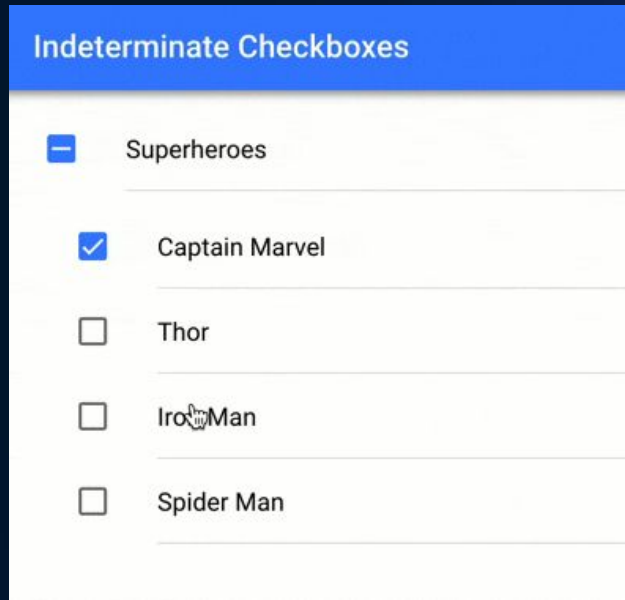
User interface **pseudo-class** selectors

:checked – radio or checkbox

- elements that have the **checked** attribute or have been **selected by the user**

:indeterminate – radio or checkbox

- elements that are in an intermediate state, **without** the checked attribute or **not selected by the user**



The screenshot shows a web form with a blue header titled "Indeterminate Checkboxes". Below the header is a section titled "Superheroes" with a minus icon to its left. Under this section, there is a list of four superheroes, each with a checkbox to its left:

- ☒ Captain Marvel
- ☐ Thor
- ☐ Iron Man
- ☐ Spider Man

User interface **pseudo-class** selectors

- **:in-range** – highlights an element when the user's input falls within the **specified range**
-

- **:out-of-range** – highlights an element when the user's input falls **outside** the specified range
-

- ```
input:in-range { }
<input min="1" max="10">
```
-

# User interface **pseudo-class** selectors

- **:read-only** – targets elements with the **readonly attribute** (indicating that the user cannot change the element's value)
- **:read-write** – targets elements that **do not have the readonly attribute**

Read-Only Input

This can only be copied



# User interface **pseudo-class** selectors

- **:target** – applies styles to elements that are referenced by a fragment identifier (ID) in a URL.

```
Link
```

```
<p id="one"> Target element </p>
```

```
p:target {
 background-color: red;
 color: white;
}
```

[Change 1st line.](#)

testing the target selector

# User interface **pseudo-class** selectors

- **:valid** – highlights an element when the entered data meets the specified patterns
- **:invalid** – highlights an element when the entered data does not meet the specified patterns.

Valid

Label \*

Invalid

Label \*

# styling valid & invalid

Data entered correctly:

```
input:valid {border-color: green;}
```

valid

Data entered incorrectly:

```
input:invalid {border-color: red;}
```

input

**Note:** This works even if the input is empty, because an empty value is also considered invalid.

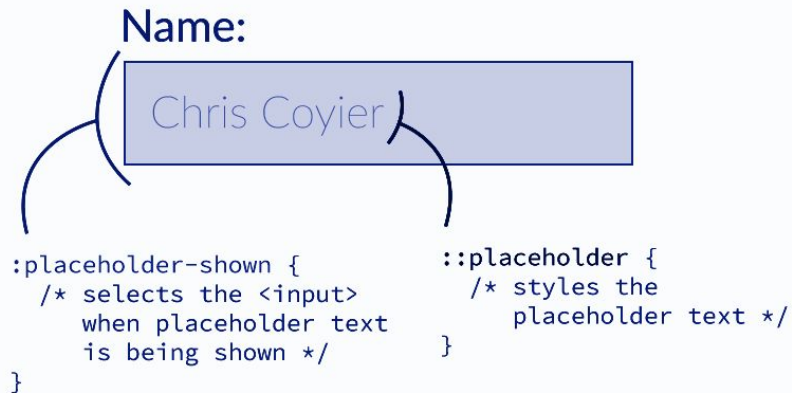


# Pseudo-elements

**::placeholder** – Styles the placeholder text inside an **input** or **textarea**, when nothing is entered into it

## Pseudo-class

**:placeholder-shown** – represents any **input** or **textarea** element that is currently displaying placeholder text.



# styling valid & invalid

Considering whether data has been entered into the input:

```
input:invalid:not(:placeholder-shown) {border-color: red;}
input:valid:not(:placeholder-shown) {border-color: green;}
```

This checks whether the placeholder is visible (it's not visible if data has been entered in the field)

# styling valid & invalid

Considering whether the input is focused:

```
input:invalid:not(:placeholder-shown):not(:focus)
{border-color: red}
```

This checks whether the placeholder is visible and whether the field is in focus. If the field is focused, the user might still be entering data, so it may be too early to validate.

# styling valid & invalid

Check after user field in field:

```
input:user-valid {border-color: green;}
```

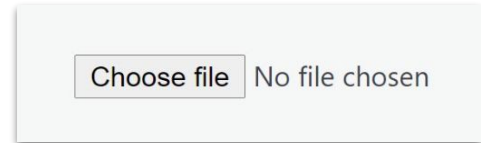
valid

Check after user field in field:

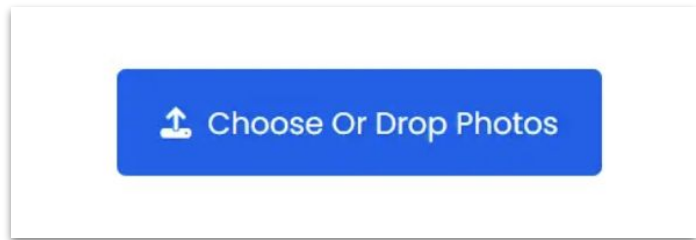
```
input:user-invalid {border-color: red;}
```

input

# styling file upload input



- Styled using pseudo-elements.
- Icons can be added using pseudo-elements.



**::file-selector-button** – Styles the button for `<input type="file">`



```
input[type="file"]::file-selector-button {}
input[type="file"]::before {}
```

# progress

Project Task 2

67% completed

↑ 13.5%

– displays the progress of long-running processes.

- Progress of filling out a registration form.
- Task completion progress.
- Progress of uploading a large file.
- Progress of course completion in online learning.
- Game progress as levels are completed.

Loading data...

26%

Installing application

In progress | About 6 minute remaining

Upload

In progress | 152 / 256 GB

# progress

```
<label for="progress-bar">Downloading</label>
```

```
<progress
```

```
 id="progress-bar"
```

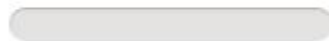
```
 value="57"
```

```
 max="100"
```

```
 title="percentage">
```

```
</progress>
```

Progress Bar - 0%



Progress Bar - 100%



Progress Bar - 57%



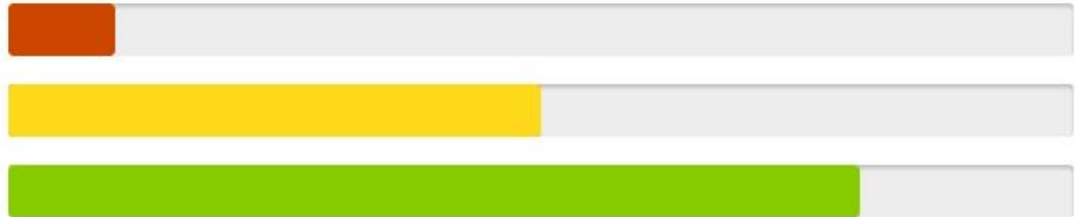
**max** – the maximum value (a number greater than 0)

**value** – the current value (a number between 0 and **max**)

# meter

– displays a scalar measurement within a known range

- Password strength
- Battery level
- Sound volume





# meter

```
<label for="vote"></label>
<meter
 id='vote'
 value='75'
 min='0'
 max='100'>
</meter>
```

**value** – current value on the scale

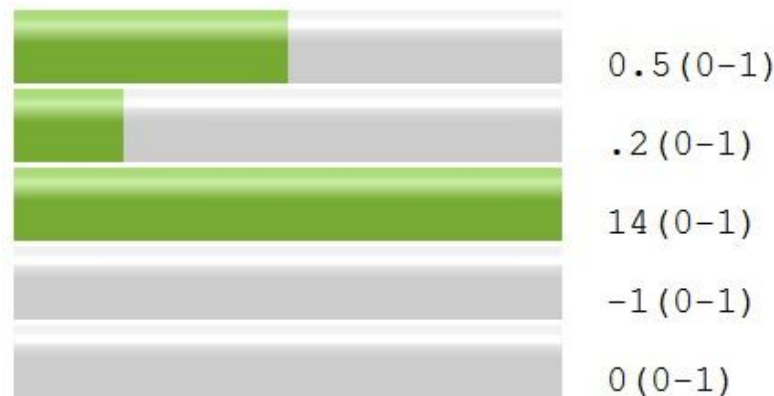
**min** – minimum value

**max** – maximum value

**low** – lower threshold value

**high** – upper threshold value

**optimum** – optimal threshold value



# appearance

**appearance: none** – resets the appearance of an element to a consistent style across all browsers and operating systems

**appearance other than none** – used to apply specific styles to elements that do not have those styles by default

**appearance: searchfield;**

The default styling applied by the browser:

**appearance: auto;**

search:

text:

date:

radio: ☐

checkbox: ☐

search:

text:

date:

radio: ☐

checkbox: ☐

# styling progress

```
progress {
 appearance: none;
}
```

```
progress::-webkit-progress-bar {
 background-color: grey;
}
```

```
progress::-webkit-progress-value {
 background-color: green;
}
```



# meter

```
meter {
 appearance: none;
}
```

```
meter::-webkit-meter-bar {
 background: none;
 /* Required to get rid of the default background property */
 background-color: grey;
}
```

```
meter::-webkit-meter-optimum-value {
 background-color: green;
}
```

Safari



Chrome



Firefox



# Quality Criteria for HTML Course

❤️ Mandatory for passing the course

💛 Required for the highest grade

💚 Optional

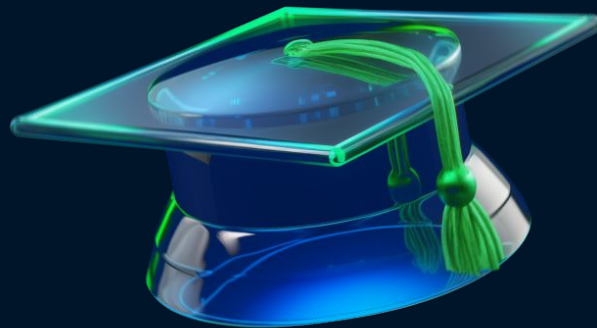
❤️ 1.19. Each form element should have a `<label>`.

💛 1.20. Resizing `<textarea>` should not break the layout.

❤️ 1.21. Basic HTML form validation should be implemented.

# Summary

1. The importance of forms on websites.
2. Components of forms.
3. How to solve styling issues with form elements.
4. How to validate form data.
5. Interactive tags outside of forms.



# Homework

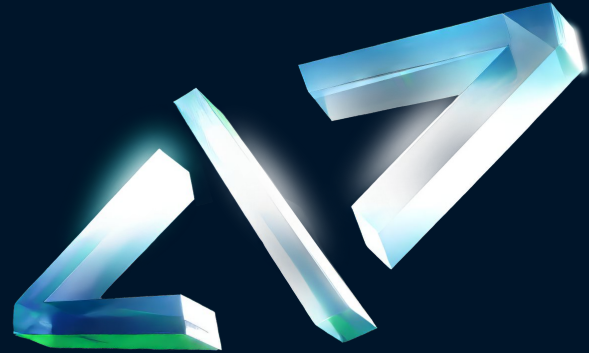
## 1. Markup and Style Forms in Binabox:

- Search in header
- Contact page
- Catalog page
  - Sidebar
  - Sort by select
  - search bar

## 2. Specify Button Types

## 3. Validate Your Forms

- required fields
- patterns



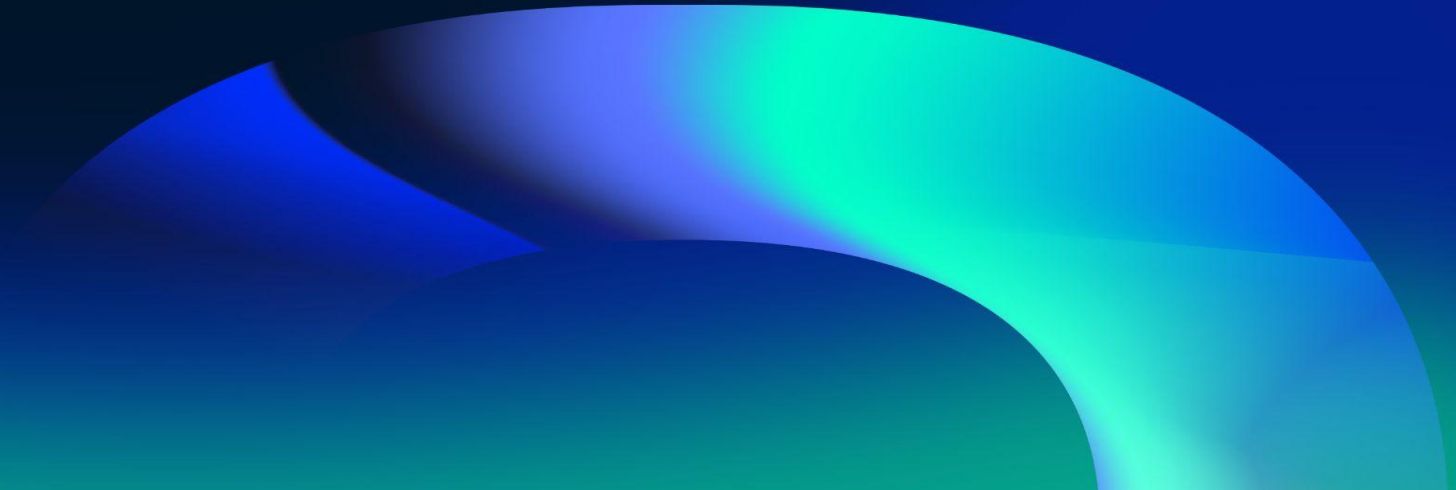
**B** Academy  
**RO**



**QUESTIONS?**



**Please fill out the feedback form**  
**It's very important for us**





**THANK YOU!**

**Have a good evening!**